

Hassan Raza

Data & Analytics Professional

Pakistan (+92) 335 5102051

razakhattak123@gmail.com

Portfolio Website: <https://hassan-raza-ktk.github.io/portfolio/>

Streamlit: <https://share.streamlit.io/user/hassan-raza-ktk>

LinkedIn: <https://www.linkedin.com/in/hassan-raza-9651b6279/>

PROFESSIONAL SUMMARY

Data & Analytics Professional with hands-on experience in Data Analysis, Machine Learning, Business Intelligence, and Software Development. Skilled in developing analytics dashboards, desktop applications, web systems, mobile applications, and AI-powered solutions using Python, SQL, Streamlit, Flask, and Firebase. Experienced in building production ready systems including face recognition attendance platforms, management information systems, Android applications, and business analytics solutions. Strong ability to transform data into actionable insights and develop technology solutions that support organizational decision-making.

CORE SKILLS

Data Analysis & Reporting: SQL (PostgreSQL, SQLite), Cloud Databases (Google Firebase, Supabase) EDA, KPIs, Dashboards (Streamlit)

Machine Learning & Modeling: Regression, Classification, NLP (TF-IDF), Anomaly Detection, Model Evaluation

Data Applications & Engineering: Python Data Apps, Streamlit Multi-page Systems, Data Modeling, Python (Tkinter), Flask, Desktop Apps

Mobile Application Development: Kotlin, Android Studio

Deployment & Delivery: Desktop Executables (.exe), Cloud Deployment, Offline-first Systems

Visualization: Plotly, Matplotlib, Seaborn

SKILLS

- Strong analytical and data interpretation skills
- Excellent communication skills
- Ability to manage priorities and cross-functional collaboration
- Python, Streamlit, Pandas, Numpy, Scikit-Learn, Seaborn, Plotly, Matplotlib, SQL, Supabase, Google Firebase
- Exploratory data analysis
- Regression analysis
- Advanced modeling
- Statistical analytics

PROJECTS

- **Face Recognition Attendance System for University of Peshawar as a Final Year Project FYP (Python, Flask, OpenCV, MySQL)**
 - Developed a face recognition-based attendance management system for Shiekh Zayed Islamic Centre classroom environments.
 - Implemented real-time facial recognition using OpenCV for automated student attendance marking.
 - Designed role-based modules for administrators, teachers, subjects, timetable management, and attendance reporting
 - Integrated MySQL database for secure storage of students, faculty, attendance records, and schedules.
 - Generated attendance analytics and reports to improve monitoring and academic record management.
- **GRE AI Coach Mobile Application (Android, Kotlin, Firebase Authentication)**
 - Designed and published an Android application to help GRE students practice Quantitative, Verbal, and Writing sections.
 - Implemented Firebase Authentication for secure user registration and login.
 - Developed quiz management, progress tracking, score analytics, onboarding screens, and AI-powered learning features.
 - Created modern Material Design UI with offline-first learning support and personalized study experience.
 - Managed complete application lifecycle including development, testing, Android App Bundle generation, and Google Play Console deployment process.
- **Khattak Tax Solutions Management System (Python, CustomTkinter, Pandas, OpenPyXL, XlsxWriter, Matplotlib)**
 - Developed a desktop-based client and taxation management system for maintaining client records, tax services, payment history, and outstanding balances.
 - Implemented secure client registration, searchable client database, payment ledger, and service tracking modules.
 - Integrated Excel reporting using OpenPyXL and XlsxWriter for automated report generation and data export.
 - Created analytical charts and business summaries using Matplotlib for monitoring collections and client activity.
 - Designed an intuitive CustomTkinter interface focused on ease of use for non-technical business staff.
 - Deployed as a standalone executable (.exe) for offline business operations.

- **Perfume Store Analytics Dashboard (Python, Streamlit, PostgreSQL, Supabase)**

Designed and developed an end-to-end analytics dashboard for a perfume retail business.

Implemented PostgreSQL database hosted on Supabase for secure cloud data storage.

Built interactive Streamlit dashboards for sales trends, product performance, revenue analysis, and customer insights.

Enabled real-time data updates and deployment-ready architecture for business decision support.

Deployed dashboard online with role-ready structure suitable for remote analytics use.

- **SMC Clinic Management Software (Python, SQLite, CustomTkinter)**

Developed a professional desktop-based Clinic Management System using Python and CustomTkinter.

The software includes secure login, patient registration, visit history tracking, medicine template management, and a configurable dashboard with real-time statistics.

Implemented SQLite for structured medical data storage and designed the system for offline clinic usage with future scalability.

Deployed as an executable (.exe) for offline use by doctor.

- **E-Commerce Website (HTML, CSS, JavaScript, Firebase)**

Designed and developed a complete e-commerce website for Lawari Food brand.

Implemented modern, responsive UI using HTML, CSS, and JavaScript.

Integrated Firebase as backend for database, authentication, and product management.

Developed admin panel features for product add/update, pricing control, and inventory handling.

Enabled order management, customer data storage, and real-time updates via Firebase cloud database.

Optimized website structure for scalability, performance, and future payment gateway integration.

- **SGCA School Management Software (Python, SQLite, Streamlit)**

Developed a desktop ERP system for a primary school with modules for student management, fee tracking, certificates, and SMS center.

Deployed as an executable (.exe) for offline use by administrators.

- **Fake News Detection ML Project (Python, Scikit-learn)**
Implemented ML models (Logistic Regression, Random Forest, SVM TF-IDF) to classify news articles.
Achieved 80% accuracy with explain ability reports for performance metrics.
- **Olympic Story: 120 Years of History (Interactive Dashboard)**
Built a Streamlit dashboard visualizing Olympic data using Python & Plotly.
Project link: <https://www.kaggle.com/hassanrazakhattak>
Live Demo: <https://olympicsdashboardbyhassanraza051.streamlit.app/>
- **Superstore Sales Dashboard (ML Forecasting & Responsiveness)**
Created a responsive dashboard with ML forecasting models (3–12 month horizon).
Live Demo: <https://superstoredashboard-byhassanraza.streamlit.app/>
- **Bellabeat Case Study: Smart Device Data Analysis**
Conducted EDA and SQL analysis for Fitbit data to generate marketing insights.
Kaggle Link: <https://www.kaggle.com/hassanrazakhattak>
- **Anomaly Detection in Telecom Networks (ML/DL Project)**
Built ML and deep learning models for traffic anomaly detection.
Kaggle Link: <https://www.kaggle.com/hassanrazakhattak>

INTERNSHIP

Machine Learning Intern – Neuro App (Remote)

Completed a structured Machine Learning internship focusing on practical implementation of core ML concepts.
Worked on NumPy and Pandas data handling, data cleaning, feature engineering, and exploratory data analysis.
Built classification and regression models using Scikit-learn and evaluated them using standard metrics.
Developed and documented weekly ML tasks with GitHub repositories, screenshots, and README files.
Gained hands-on experience in data visualization, model persistence, and end-to-end ML workflows.

[Internship Repositories](#)

EDUCATION

BS Computer Science- From University of Peshawar

CERTIFICATES

Google Career Certificate in Advanced Data Analytics

Google Career Certificate in Data Analytics

Project Management by Google

Machine Learning with Python by IBM

Python for Everybody by University of Michigan

Using Databases with Python by University of Michigan

LANGUAGES

English, Urdu, Pashtu